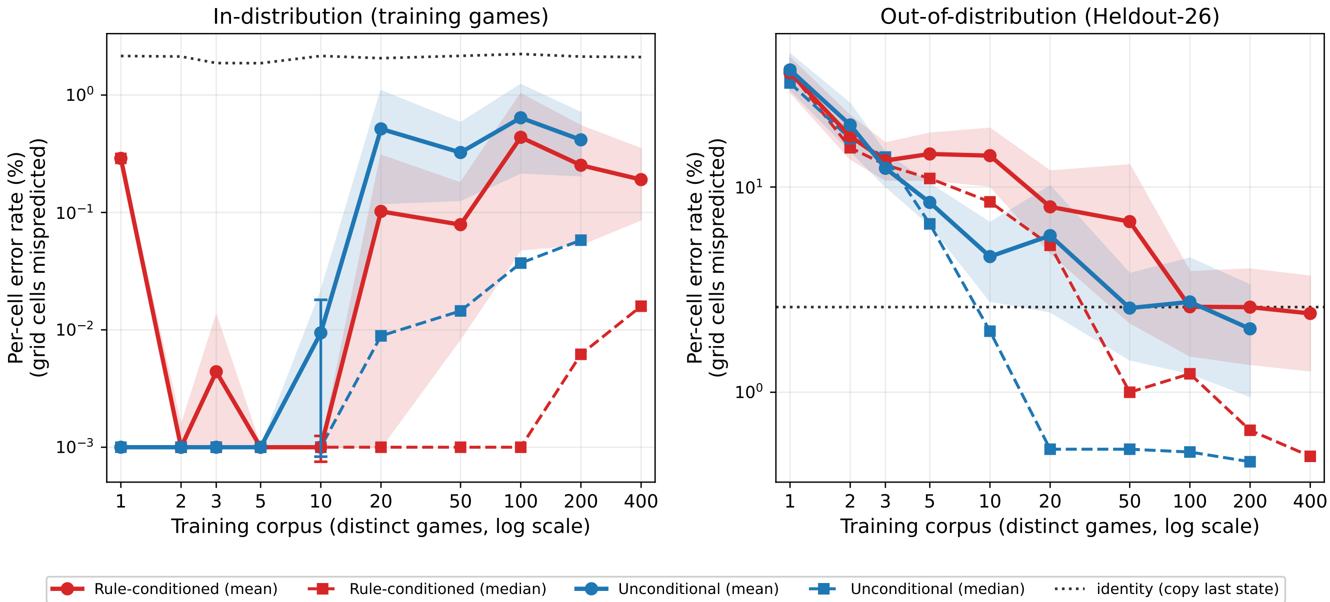

In-distribution (training games)
Out-of-distribution (Heldout-26)
Per-cell error rate (%)
(grid cells mispredicted)
10^0
10^{-1}
10^{-2}
10^{-3}
10^1
10^0
1 2 3 5 10 20 50 100 200 400
Training corpus (distinct games, log scale)
Rule-conditioned (mean)
Rule-conditioned (median)
Unconditional (mean)
Unconditional (median)
identity (copy last state)

Data-scaling of the NCA world model vs. training-corpus size. Each panel plots per-cell error rate (the fraction of grid cells whose predicted object-stack differs from the ground truth, averaged over teacher-forced rollout steps and over games) against the number of distinct training games, selected in ascending rule-count order from the token-deduplicated pool (the `n_per_rule` sweep), on log–log axes. *Left:* in-distribution, aggregated over each corpus's own training games. *Right:* out-of-distribution, aggregated over the fixed Heldout-26 set of unseen games. Two parameter-matched models are compared: the rule-conditioned model (n_h=256 plus the text-encoder/decoder/slot stack, ≈15.97M parameters) and an unconditional model given slightly more capacity (n_h=288, ≈16.61M), so any conditioning benefit cannot be attributed to parameter count. Solid lines are the mean over games (or heldout pairs); dashed lines are the median. The shaded band is a bootstrap 95% confidence interval of the mean *across games* (left) or *across heldout pairs* (right). The capped vertical error bars show the *across-seed* spread (min–max of the per-seed mean), drawn only at corpus sizes with at least two training seeds; they measure run-to-run reproducibility and are distinct from the across-game band. The dotted black reference is the identity (copy-last-state) baseline—per-corpus state churn in-distribution (≈2%) and 2.60% on Heldout-26. In-distribution error stays near zero and dilutes only mildly as the corpus widens at fixed capacity, with the rule-conditioned mean degrading more slowly than the unconditional one; out-of-distribution error falls toward the identity baseline as the corpus grows.